Project Definition

The project is to develop a System-On-Board (SOM) based on Analog Devices BF548 Digital Signal Processor (DSP). The component diagram of the card is shown on Figure 1.

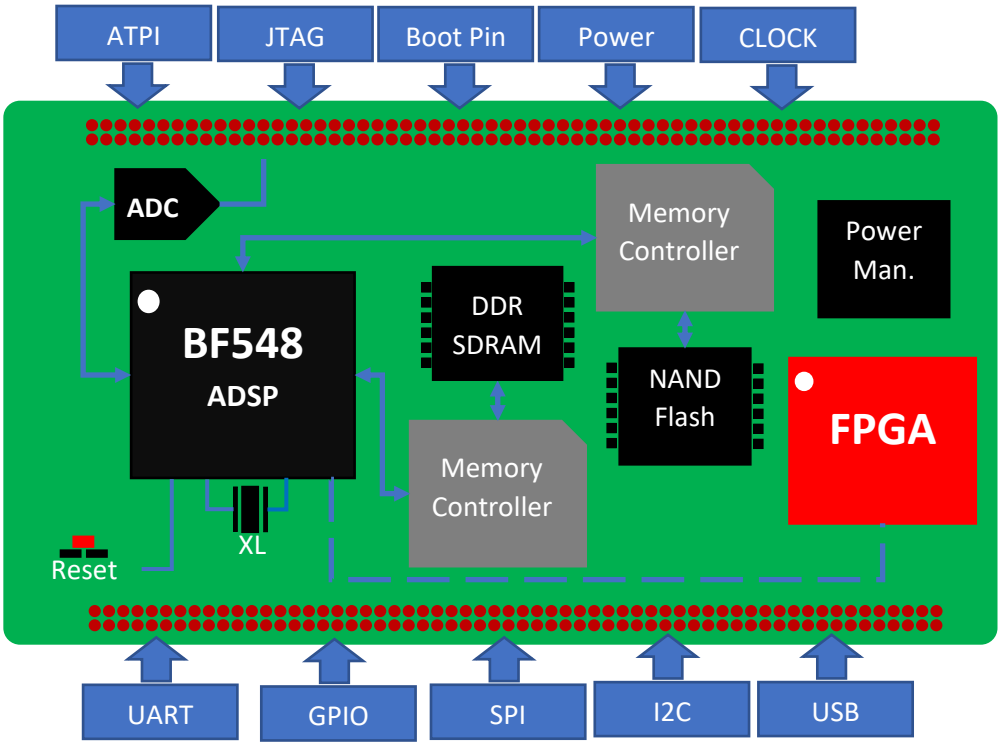


Figure 1 System block diagram

MILESTONES

In the *first* milestone, inspiring from BLUE TECHNIX's CM-BF548 card, a card with the following technical specifications is envisioned:

- Support BF548
- Support DDR RAM (the part number is flexible to choose)
- Support NAND flash (the part number is flexible to choose)
- Support Reset button

- Support an external ADC (minimum requirements: 8 channels, 10khz)
- Support Power Management IC (PMIC)

Design specifications of the first milestone:

- To provide an extra space on the board, the card size may not be the same as the CM-BF548
- The pin out of the card may differ from CM-BF548. The final complete pin out map would be provided in the project output reports.
- On board components such as ADC, RAM and Flash would be checked for meeting the project requirements.
- In order to simplify the circuit complexity some unused peripherals of the BF548 may not get connected to the headers (such as MXVR, audio, etc.) and instead of them new pin (ADC pins) may added.
- An external +3.3V is fed into the card.
- All UART communication-channels are TTL.

In the *second* milestone, an FPGA is planned to mount on the card. In this milestone the FPGA and DSP may directly connect to each other. This new requirement changes the power specifications of the card and may require revisiting the power management specification.

PLANNING

The first iteration is planned to be implemented in two phases. In the phase one, except power management unit, all components are inserted on the card. In this phase the card is powered by external 3.3V. In the second phase, the power management unit is also added and then evaluated. In this way, all power related issues are debugged separately.

The second milestone is also planned in two phases, first an FPGA is added, and no changes are made on the card. If any issue such as interference, EMC, power line malfunction, clock mismatch, etc. happened, then the card is revisited for next phase, otherwise the development is the project is finished.

TOOLS & ENVIRONMENT

All designs would be based on Altium designer 20 and the outputs are transferred over a GitHub repository.

BUDGET

Item	Cost
Design circuit which meets the first milestone requirements + training	5000 USD
Technical consultant for second milestone	20 USD / hour